

EXECUTIVE SUMMARY

Salifort Motors Employee Retention Project

Slide 1: Project Overview

Objective: To design a data-driven machine learning model that predicts employee turnover at Salifort Motors, enabling HR to proactively retain highly skilled and valuable talent in a competitive market.

Methodology: Analyzed an HR dataset of 14,999 employees. Conducted data cleaning, structural EDA, and deployed ensemble machine learning algorithms (XGBoost and Random Forest) to identify the core drivers of attrition.

Slide 2: Results of Analysis

- **The "Extreme Edges" Phenomenon:** Turnover is heavily concentrated at the extreme ends of the operational spectrum. Employees with excessive workloads or very few projects leave at high rates, while the "middle core" remains stable.
- **Top Drivers of Churn:** The model conclusively identified that operational bandwidth drives turnover. The most critical predictive features are *average_monthly_hours* (Importance Score: 905.0), *satisfaction_level* (791.0), and *last_evaluation* (726.0).
- **Low Impact Factors:** Variables such as baseline salary and specific departments showed minimal predictive importance compared to daily workload.

Slide 3: Model Benefits & Limitations

- **Benefits:** The chosen XGBoost model is highly accurate (99% overall accuracy). It was specifically selected for its computational speed with big data and its strong resistance to overfitting on unseen future data. It also inherently handles outliers (like veteran employees) without requiring destructive data dropping.
- **Limitations:** The model currently relies on static, point-in-time data. It cannot capture the chronological growth of an employee's skills or changing market conditions without periodic retraining.

Slide 4: Recommendations & Next Steps

- **Workload Rebalancing:** HR must strictly monitor *average_monthly_hours* to prevent burnout among high-performing employees who are burdened with too many projects.
- **Implement "Employee Portfolios":** Develop a continuous tracking system to evaluate employee skill acquisition and performance every 6 months. This allows HR to proactively foster loyalty as employees become more valuable to competitors.
- **Pilot Phase Deployment:** Launch a shadow deployment of the XGBoost model for new hires and current employees. This allows HR to validate predictions in real-time and make rapid, short-term model calibrations without disrupting existing workflows.